Module-1: Overview on Automation & Selenium

1. Introduction to Automation
2. Challenges in Manual Testing
3. Overcoming Challenges with Automation
4. Importance of Programming in Automation
5. Introduction to Selenium
6. Selenium Components
7. Advantages and Disadvantages of Selenium WebDriver

Module-2: Python Programming for Selenium

1. **Download and Install Python**
    a. Overview of Python
    b. Downloading Python from the official website
    c. Installing Python
2. **Setup Python Environment**
    a. Setting up Python Path in Environment Variables
    b. Verifying Python Installation
    c. Introduction to pip (Python Package Installer)
3. **How to use PyCharm**
    a. Installing PyCharm IDE
    b. Setting up a new project in PyCharm
    c. Navigating the PyCharm interface
4. **Basic Python Program**
    a. Writing the first Python script
    b. Understanding the Python file structure
    c. Running the Python script
5. **Run a Python Program**
    a. Running Python programs from the command line
    b. Running Python scripts in PyCharm
6. **Understanding Console Output**
    a. Using the print() function
    b. Formatting console output

7. **Python Variables and Data Types**
    a. Variable declaration and initialization
    b. Different data types (int, float, string, boolean)
    c. Type casting
8. **Python Operators**
    a. Arithmetic operators
    b. Comparison operators
    c. Logical operators
    d. Assignment operators
9. **Conditional Statements**
    a. if statement
    b. if-else statement
    c. elif statement
    d. Nested if statements
10. **Looping Statements**
    a. for loop
    b. while loop
    c. Nested loops
    d. Loop control statements (break, continue, pass)
11. **Lists and Tuple**
    a. Creating and accessing lists
    b. List methods
    c. Tuples and their characteristics
    d. Tuple operations
12. **String**
    a. String declaration
    b. String methods and operations
    c. String formatting
13. **Dictionary and Set**
    a. Creating and accessing dictionaries
    b. Dictionary methods
    c. Creating and accessing sets
    d. Set operations and methods

Module 3: Python Object Oriented Programming

1. **Classes and Objects in Python**
    a. Defining a class
    b. Creating objects from a class
    c. Instance variables and methods
2. **Functions and Methods in Python**
    a. Defining functions inside and outside of a class
    b. Calling methods
3. **Passing Parameters to Methods**
    a. Positional arguments
    b. Keyword arguments
    c. Default parameters
4. **Python Constructor**
    a. Understanding the __init__ method
    b. Creating constructors
    c. Constructor overloading with default arguments
5. **Method Overloading**
    a. Concept of method overloading in Python
    b. Using default arguments for method overloading
6. **Constructor Overloading**
    a. Simulating constructor overloading with default arguments
7. **Static Variables and Methods**
    a. Defining static variables
    b. Defining static methods using @staticmethod decorator
8. **Python Inheritance**
    a. Single inheritance
    b. Multiple inheritance
    c. Multilevel inheritance
    d. Hierarchical inheritance

9. **Method Overriding**
    a. Concept of method overriding
    b. Using super() to call parent class methods
10. **Python Packages**
    a. Creating and using packages
    b. Importing modules from packages
    c. Managing namespaces
11. **Exception Handling**
    a. Understanding exceptions
    b. Try, except, else, and finally blocks
    c. Raising exceptions
    d. Custom exceptions

Module-4: Selenium WebDriver (Latest Version)

1. **Introduction to WebDriver and Architecture**
    a. Overview of Selenium WebDriver
    b. Architecture of Selenium WebDriver
2. **Environment Setup & WebDriver Configuration**
    a. Setting up WebDriver in PyChamp
3. **How to Create Automated Test Case in WebDriver?**
    a. Setting Up a Basic Test Case Structure
    b. Writing and Running Simple WebDriver Test Cases
4. **Execute Test Case on Multiple Browsers**
    a. Cross-Browser Testing
    b. Running Tests on Different Browsers
5. **Understanding HTML Tags and Attributes**
    a. HTML Basics for Selenium Testers
    b. Locating Elements Using HTML Tags and Attributes
6. **Basic Locators**
    a. Name Locator
    b. ID Locator
    c. LinkText Locator
    d. PartialLinkText Locator

7. **Advanced Locators**
    a. CSS Locators
    b. XPath Locators
    c. XPath Axes
8. **Relative Locators**
    a. .above, .below, .to_left_of, .to_right_of, .near
9. **How to Use SelectorsHub & Features of Selectors Hub**
    a. Overview of SelectorsHub
    b. Features and Benefits
10. **WebDriver Methods**
    a. Opening Page URL
    b. Verifying Page Title
    c. Getting Page Source
    d. Getting Page URL
    e. Conditional Methods
        i. Visibility of WebElements
        ii. Enable/Disable Status of Web Elements
11. **Waits**
    a. Implicit Waits
    b. Explicit Waits
    c. Page Load Time
12. **Navigation Methods**
    a. Navigating Back & Forward
    b. Refreshing Page
13. **Browser Commands**
    a. close() and quit()
14. **Working with Web Elements**
    a. Handling Input Boxes
    b. Handling Checkboxes & Radio Buttons
    c. Handling Links
15. **Handling Alerts/Popups**
    a. Handling Alert Boxes
    b. Handling Confirmation Boxes

c. Handling Prompt Boxes

16. **Handling Iframes & Inner Iframes**
    a. Switching to Iframes
    b. Working with Nested Iframes
17. **Handling Drop Downs**
    a. Select type Drop down
    b. Bootstrap Drop Downs
    c. Multi Select Drop Down
    d. Auto Suggest Drop Down
18. **Working with Browser Windows**
    a. Switching Between Browser Windows
    b. Handling Tabs and Windows
19. **Handling Static & Dynamic Web Tables**
    a. Strategies for Handling Tables
    b. Handling Static Tables
    c. Handling Dynamic Tables
    d. Pagination Tables
20. **Handling Date Pickers**
    a. Strategies for Handling Date Pickers
21. **Handling Mouse Actions**
    a. Using Actions Class
    b. Right Click, Double Click, Mouse Over
    c. Drag and Drop Actions
22. **Handling Keyboard Actions and Shortcuts**
    a. Using Actions Class for Keyboard Actions
23. **Executing Javascript code in Selenium**
24. **Scrolling Web Page**
    a. Scrolling Up and Down the Page
25. **Zoom In & Zoom Out, Maximize, Minimize Web Page**
    a. Managing Browser Window Size
26. **Opening Links in New Tab**
    a. Handling New Tabs
27. **Working with Tabs and Windows**

a. Switching Between Tabs and Windows

28. **Capturing Screenshots**
    a. Taking Screenshots in Selenium
29. **Working with Cookies**
    a. Managing Cookies in Selenium
30. **Headless Browser Testing**
    a. Overview of Headless Testing
    b. Advantages and Disadvantages
31. **Browser Settings with ChromeOptions**
    a. Customizing Chrome Browser Settings
32. **Run Tests in Incognito Mode**
    a. Running Tests in Incognito/Private Mode
33. **How to Handle Upload Files (Single and Multiple)**
    a. Uploading Files in Selenium
34. **Handling Broken Links**
    a. What is Broken link?
    b. How to find Broken links in web page
35. **Handling shadowDOM, SVG Elements**
    a. Strategies for Handling shadowDOM and SVG Elements
36. **Working with Excel Files Using Apache POI API**
    a. Reading and Writing Excel Files
37. **Data-Driven Testing**
    a. Strategies for Data-Driven Testing
    b. Implementing Data-Driven Tests in Selenium
38. **Reading Properties File**

Module-5: PyTest

1. **Introduction to PyTest**
    a. Overview of PyTest
    b. Key Features and Benefits
2. **PyTest Installation**
    a. Installing PyTest via pip
    b. Verifying the Installation
3. **PyTest Naming Conventions**

   a. Standard Naming Conventions for Test Files and Test Functions
   b. Importance of Naming Conventions
4. **PyTest Test Case**
   a. Writing Simple Test Cases
   b. Understanding Assertion Methods in PyTest
5. **PyTest Fixtures**
   a. Introduction to Fixtures
   b. Creating and Using Fixtures
   c. Scope of Fixtures (Function, Class, Module, Session)
6. **How to Use Common Fixtures in Multiple Modules**
   a. Sharing Fixtures Across Multiple Test Modules
   b. Using conftest.py for Common Fixtures
7. **Run All the Modules in a Package**
   a. Running Tests from All Modules in a Package
   b. Configuring PyTest to Discover and Run Tests
8. **Run Specific Module from a Package**
   a. Command to Run Specific Test Modules
   b. Use Cases for Running Specific Modules
9. **Run Specific Test Method from a Module**
   a. Running Individual Test Methods
   b. Specifying Test Methods in the Command Line
10. **Grouping Tests**
   a. Organizing Tests into Groups
   b. Using Markers to Group Tests
11. **Skipping Tests**
   a. Skipping Tests Conditionally
   b. Using pytest.mark.skip and pytest.mark.skipif
12. **Ordering Tests**
   a. Controlling the Order of Test Execution
   b. Using pytest-ordering Plugin
13. **Parallel Testing using PyTest xdist-plugin**
   a. Introduction to Parallel Testing
   b. Installing and Using pytest-xdist for Parallel Execution

c. Configuring Number of Parallel Processes

14. **Parameterization**
    a. Parameterizing Tests for Different Input Data
    b. Using pytest.mark.parametrize for Test Cases
15. **Command Line Arguments**
    a. Common Command Line Options
    b. Custom Command Line Arguments
16. **Structure Tests in a Test Class**
    a. Organizing Tests in Classes
    b. Setup and Teardown Methods in Test Classes
17. **Return a Value from Fixture**
    a. Returning Values from Fixtures
    b. Using the Return Value in Test Functions
18. **PyTest HTML Report Generation**
    a. Generating HTML Reports
    b. Using pytest-html Plugin
    c. Customizing the HTML Report

Module-6: Extent Reports, Logging & Page Object Model Pattern

Generating Extent Reports
Python Logging module for Logging
Page Object Model (POM) Pattern
Creating Page Object Classes using Page Factory

Module-9: Introduction to Automation Frameworks

Introduction to Automation Frameworks
Types of Frameworks
Phases in Designing Framework
Choosing Test Cases for Automation

Module-10: Hybrid Framework with POM (Page Object Model) – Project

1. Creating New Project and add required plugins
2. Creating Folder Structure
3. Creating Page Objects and Object Repository
4. Creating Utility files

5. Generating Reports
6. Generating Logs using python Logging module
7. Creating Configuration Files
8. Creating Automation Test Cases
9. Creating Data-Driven Test Cases
10. Creating Database Test Cases
11. Executing Test Scripts
12. Emailing Test Reports
13. Executing Test Scripts through Command Line

Module-11: CI & CD Concepts(Git, Github)

1. Version Control System Overview
2. Git and GitHub
3. Git Commands

Module- 12- AI with GitHub Copilot

1. GHCP Configuration
2. GHCP Prompt Engineer Messages

Module -13- MySQL

1. MYSQL concepts
2. MYSQL Configuration
3. MYSQL Query statments